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AI-enhanced Business Intelligence for decision-making

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The rapid expansion of data and growing market uncertainty have positioned Artificial Intelligence (AI)-enhanced Business Intelligence (BI) as a key enabler of effective decision-making. The primary objective of this study was to identify the synergy between BI and AI, including the benefits and challenges arising from their integration into decision-making processes, as well as to analyze how managers perceive this technological convergence. To achieve this goal, a qualitative approach was employed, enabling an in-depth exploration of the participants' experiences and perspectives. The findings indicate that AI-powered BI enhances decision-making speed, improves analytical accuracy, automates insights delivery, and increases strategic agility. However, major barriers to this synergy include high implementation costs, data quality issues, and a shortage of highly qualified specialists. The study confirms that the integration of AI and BI significantly enhances organizational performance and supports data-driven, future-oriented decision-making.

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Contemporary organizations operate in an environment marked by an overwhelming volume of data and rapidly evolving business dynamics. In this context, the ability to transform data into actionable insights has become a strategic imperative. Business Intelligence (BI), which has traditionally functioned as a tool for reporting and analysing historical data, is undergoing a profound transformation. It is evolving into an intelligent, Artificial Intelligence (AI)-enhanced system capable of supporting advanced predictive and prescriptive analytics. The growing incorporation of AI into BI systems is enabling the emergence of AI-enhanced Business Intelligence, where technologies such as

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Machine Learning (ML) and Natural Language Processing (NLP) automate data interpretation and support decision-making processes through real-time, data-driven recommendations [1], [2], [3]. This synergy redefines the role of BI - from passive reporting to active participation in strategic and operational decision-making.

Although the potential of AI in BI has been widely acknowledged in the literature, existing studies often focus on technological architectures or theoretical models. There is a noticeable lack of empirical research examining how organizations actually implement AI-enhanced BI and how it influences real-world decision-making. In particular, qualitative insights into managerial experiences, perceived benefits, implementation challenges, and organizational impacts remain limited. This research gap is especially relevant in the context of the understanding the practical synergy between AI and BI within decision-making environments.

The primary objective of this study is to identify the synergy effect between BI and AI, including the benefits and barriers resulting from their integration into decision-making processes, as well as to analyze how managers perceive this synergy. To achieve this goal, a qualitative research approach was employed, allowing for an in-depth exploration of the participants' experiences and perspectives. The use of in-depth interviews (IDIs) was driven by the need to obtain rich, context-sensitive insights into the practical aspects of implementing and utilizing AI-driven BI systems. Qualitative interviews are particularly well-suited for exploring emerging phenomena and understanding complex the decision-making environments.

The study was conducted among 20 organizations representing three economic sectors (services, trade, and manufacturing) where BI and AI are actively used to support business decision-making. The respondents included managers, data analysts, and AI/BI specialists, providing a diverse range of perspectives on the applications, advantages, and challenges associated with the integration of these technologies.

The paper is structured as follows. Following the introduction, which outlines the motivation, research gap, and the methods, the next section presents a literature review detailing the evolution of BI and its integration with AI in the context of decision-making. This is followed by a comprehensive discussion of the research methodology, including sample selection, data collection, and analytical procedures. The subsequent section presents and discusses the research findings, highlighting the key insights regarding the benefits and barriers of AI-enhanced BI implementation. The paper concludes with a summary of the study's contributions, limitations, and recommendations for future research.

1. From Reactive to Augmented Analytics

1.1. Business Intelligence

Numerous scholars agree that BI is not merely a technology but above all, a comprehensive information management paradigm. It encompasses processes, technologies, and tools that enable the collection, analysis, and interpretation of data to support business decision-making [3], [4], [5], [6].

BI plays a crucial role in transforming raw data into actionable knowledge, enhancing operational efficiency, strengthening competitive advantage, and facilitating more informed strategic decision-making. It integrates multiple dimensions of data analysis, particularly those essential for managerial decision support, including [7], [8], [9]:

- data collection from multiple sources, including transactional systems, databases, and social media platforms;
- data processing and integration, including data cleansing, transformation, and consolidation within data warehouses;
- data analysis and exploration, encompassing advanced techniques such as data mining, predictive analytics, and scenario modeling;
- data visualization and reporting, providing managers with clear and interactive reports as well as executive dashboards.

In the literature, considerable attention is devoted to the distinctive features of BI systems that define their evolving functionality. The key elements highlighted include [10], [11], [3], [12]:

- beyond data collection -emphasis on analysis – while traditional IT systems primarily record and store data, BI focuses on data analysis and interpretation;
- support for all management levels - BI provides both high-level strategic insights for executives and detailed operational reports for business units;

- integration with multiple systems - BI consolidates data from ERP, CRM, databases, e-commerce platforms, and social media, enabling a holistic view of the organization;
- predictive and prescriptive analytics – Advanced BI systems can forecast future trends and recommend actions, making them significantly more effective than traditional reporting systems.

It is also worth noting the specific architecture of BI systems compared to other IT systems. The key components of this architecture include [3], [13], [14], [15]: (1) Data Warehouses – central repositories that store integrated data from various sources; (2) ETL (Extract, Transform, Load) processes – mechanisms for extracting, transforming, and loading data into the data warehouse; (3) OLAP (Online Analytical Processing) analysis – enabling multidimensional data analysis in real time; (4) Data Mining – techniques for discovering hidden patterns and trends within data; (5) Data visualization and dashboards – tools that facilitate the interactive presentation of analytical results.

A key aspect from the perspective of organizational functioning is the set of benefits that companies can achieve by leveraging BI systems. Research findings confirm that the most frequently cited advantages include [16], [17], [11], [9]:

- increased operational efficiency – automation of reporting and real-time access to data enable organizations to respond more swiftly to changes;
- enhanced decision-making – BI provides managers with reliable information, reducing the risk of erroneous decisions;
- greater competitiveness – market analysis, customer behavior insights, and trend identification allow firms to adapt their strategies to a dynamically evolving environment;
- cost optimization – BI facilitates the identification of areas where cost savings are possible, improving resource management;
- fraud and anomaly detection – real-time data analysis enables the rapid identification of irregularities in financial and business operations.

1.2. Evolution of Business Intelligence Systems

Analyzing the subject literature, it is evident that (BI has undergone significant evolution - from simple reporting systems to advanced analytical tools. Typically, three main generations of BI are distinguished [18], [19], [20]:

- BI 1.0 - focus on reporting and historical data analysis. BI primarily relies on databases and static reports that provide insights into past organizational activities;
- BI 2.0 - emphasis on interactive analysis and integration with enterprise management systems. BI becomes more accessible to business users, enabling real-time data analysis;
- BI 3.0 and BI 4.0 - BI systems leverage artificial intelligence (AI), machine learning, and Big Data. They enable predictive and prescriptive analytics, allowing for automatic pattern detection in data and the recommendation of optimal actions.

Davenport (2018), in his article "From Analytics to Artificial Intelligence," presents a similar evolution of business analytics corresponding to the previously proposed classification. The author identifies four key stages that reflect the increasing role of AI in data analysis. These stages are [21]:

1. Analytics 1.0 - Traditional BI Analytics (1970s–1990s), characterized by:
 - reliance on data warehouses, Online Analytical Processing (OLAP) systems, and static reports;
 - traditional data analysis using simple query languages to support internal managerial decisions;
 - predominant use of historical data, limiting the ability to predict future trends.
2. Analytics 2.0 - The Era of Big Data (post-2000). Analytics 2.0 is characterized by the following developments:
 - the introduction of modern data storage and processing technologies, such as Hadoop and NoSQL;
 - the collection and analysis of vast amounts of data from multiple sources, including social media and the Internet of Things (IoT);
 - automated analysis and predictive modeling enabled by machine learning algorithms.
3. Analytics 3.0 - Digital Transformation and Intelligent Analytical Systems (post-2010). This phase of analytics is distinguished by:
 - the integration of BI with artificial intelligence systems across various industries;

- the implementation of advanced predictive and prescriptive techniques, allowing for dynamic business decision recommendations;
 - the automation of analytical processes -leveraging AI and ML to enhance analytical efficiency and reduce the need for human intervention.
4. Analytics 4.0 - The Era of AI-Driven BI and Augmented Analytics (post-2020). This latest stage of analytics involves the following advancements:
- BI evolves toward Augmented Analytics, incorporating AI and NLP to enhance analytical capabilities;
 - AI autonomously generates business insights, identifies anomalies, and provides real-time decision recommendations;
 - data is analyzed autonomously, and systems learn from user interactions, reducing the need for advanced analytical expertise among managers.

In conclusion, BI is an information management concept primarily aimed at transforming data into strategic knowledge. Preliminary research analyses indicate that new opportunities arise with proactive analytics an approach that integrates the advancements of traditional BI systems with AI.

1.3. Integration of Business Intelligence and Artificial Intelligence in Organizational Management

The literature indicates that AI is playing an increasingly significant role in organizational management by automating data analysis and supporting decision-making processes. It is believed that the integration of AI with BI enables the transformation of analytics from reactive to proactive, where systems autonomously detect patterns and suggest optimal strategic actions. This integration leads to the evolution of decision-making models, replacing traditional approaches based on historical data analysis with dynamic systems that leverage predictive and prescriptive analytics, as well as ML and NLP technologies [22], [23], [24], [25].

It is worth recalling that in the traditional decision-making process, decisions were made based on managers' intuition and static reports generated by BI systems. In other words, decision-making relied on the following models [26], [27]:

- the rational model, which assumes that decision-makers have full access to information and choose the best option based on logical data analysis;
- the bounded rationality model, which suggests that managers cannot process all available information and therefore rely on heuristics and simplified analytical methods;
- the intuitive model, which assumes that decisions are made based on experience and subjective assessment of the situation, potentially leading to cognitive biases.

The application of proactive analytics and, subsequently, augmented analytics, which is a consequence of BI and AI integration, requires the adoption of new tools and technologies. The most important ones include [28], [29], [30], [31]:

- Machine Learning - algorithms that analyze data, detect patterns, and generate predictions (e.g., regression models, neural networks);
- Deep Learning - used for analyzing complex data such as images, sound, and natural language;
- Expert systems - AI based on rules and knowledge bases, applied in areas such as medical diagnostics and financial risk analysis;
- Natural Language Processing - technologies that enable human interaction with BI systems through natural language, such as chatbots and semantic search;
- AI-driven decision automation - AI in BI not only analyzes data but also makes decisions, eliminating human errors and accelerating operational processes.

The areas that can benefit most from BI and AI integration include [32], [33]:

- real-time decision-making - AI analyzes data on the fly, providing dynamic strategic recommendations;
- automated data analysis - AI detects correlations and hidden patterns that previously required manual analysis;
- new data management models - AI integration enhances the efficiency of Big Data processing and storage;
- hyperautomation - the combination of AI, Robotic Process Automation (RPA), and BI to fully automate decision-making processes.

2. Research method

The primary objective of this study was to investigate how the integration of Business Intelligence and Artificial Intelligence impacts decision-making processes within organizations, the benefits and barriers it entails, and the key areas where it can be applied. The research method employed in this work was In-Depth Interviews (IDIs), which provided detailed insights into companies' experiences regarding the influence of AI and BI integration on decision-making processes.

The study was conducted between November and December 2024 and involved a sample of 20 companies representing three economic sectors:

- service sector – including companies from IT, finance, logistics, and consulting industries;
- trade sector – covering retail, wholesale, and e-commerce;
- manufacturing sector – including production, energy, and automotive industries.

The respondents in this study included managers, data analysts, and AI/BI specialists directly involved in data analysis and decision-making processes. The key criteria for selecting the surveyed organizations were:

- organization size (minimum 50 employees);
- duration of AI and BI usage (at least 6 months);
- active utilization of AI and BI systems in decision-making processes.

The study aimed to obtain diverse responses, focusing on:

- how organizations perceive the synergy between AI and BI;
- the main objectives of AI and BI usage in decision-making processes;
- the functional areas where AI and BI integration is applied;
- the potential benefits of AI and BI integration for organizations;
- the challenges and barriers associated with implementing AI and BI integration.

All data analyses and visualizations were conducted using Power BI, employing two methods: Word Cloud Analysis and Treemap Visualization. Word Cloud Analysis is a text analysis technique that visually represents the frequency of words in a dataset. Words that appear more frequently are displayed in a larger font, making it easier to identify key concepts and dominant themes in the analyzed text. Treemap Visualization is a hierarchical data visualization technique that represents relationships between categories using rectangles of varying sizes and colors. The area of each rectangle is proportional to the value of a given category, allowing for a quick understanding of the structure and dominant elements in the dataset. [34], [35], [36].

3. Findings and discussion

The conducted keyword analysis on the synergy between AI and BI in the context of decision-making processes highlights significant benefits resulting from their integration. It can be inferred that a critical aspect for the examined organizations is the automation of analytical processes (15 indications), which enhances both efficiency and the accuracy of decision-making. AI enhances BI through advanced data analysis (18 indications), enabling the identification of hidden patterns and the forecasting of market trends. The interaction between these technologies contributes to business process optimization (14 indications), improving workflow dynamics and reducing the time required for information analysis. Furthermore, the integration of AI and BI facilitates better personalization of business decisions (12 indications), aligning strategies with specific market conditions. The use of artificial intelligence algorithms enhances data visualization and supports the interpretation of analytical results (12 indications), leading to greater transparency in decision-making processes. Additionally, the application of automated pattern recognition and recommendation systems strengthens corporate competitiveness (12 indications), allowing organizations to respond more rapidly to market changes (Fig. 1).

In this survey, respondents were asked to identify the most significant benefits of integrating AI and BI within organizations. The most frequently cited advantage was the increased speed of decision-making (17 indications), suggesting that AI and BI enhance data analysis and reduce the time required to select optimal strategies. Another key benefit is decision-making support (15 indications), which enables more precise, data-driven choices while minimizing the risk of errors. Respondents also highlighted anomaly detection and cost reduction (8 indications) as crucial advantages of AI and BI implementation. The ability to automatically identify irregularities in data and optimize operational costs significantly improves organizational efficiency. Less frequently mentioned benefits included decision personalization and risk prediction (4 indications), possibly reflecting differences in the level of advanced technology adoption across industries. Industry-specific analysis revealed that the retail sector derives the most significant benefits from AI and BI implementation, with the highest number of positive responses (28 indications). This may be due to the dynamic nature of the industry and the need for rapid adaptation to changing market conditions. The service (17 indications) and manufacturing sectors (17 indications) also recognize substantial benefits from adopting modern technologies in decision-making processes, although to a slightly lesser extent than retail (Fig. 4).

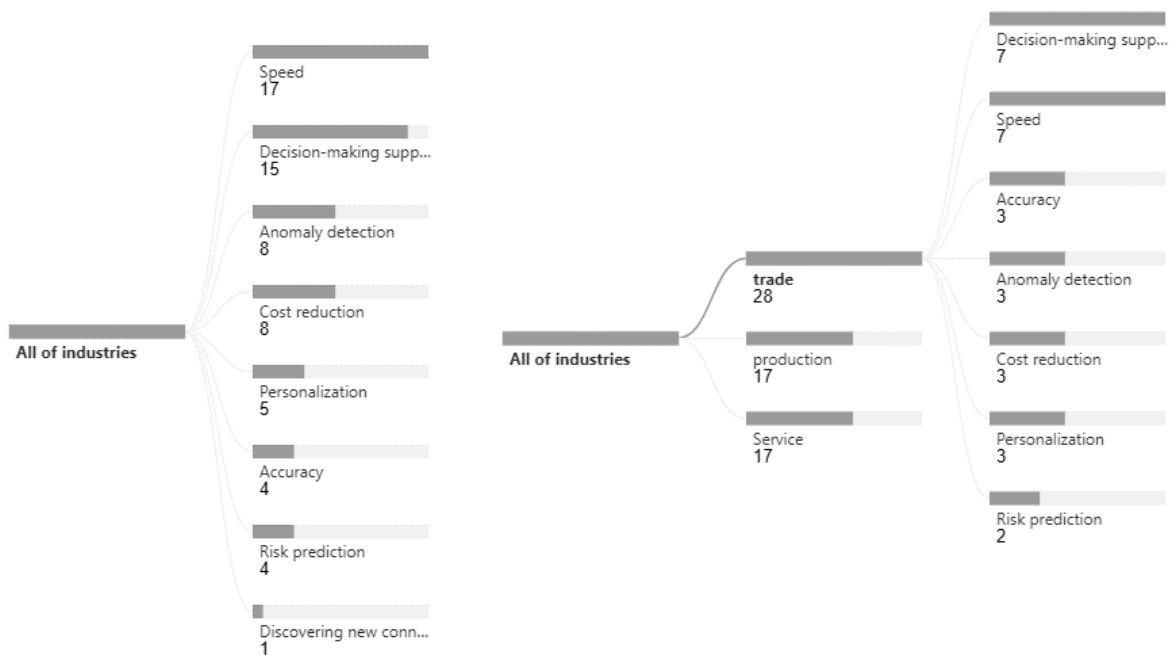


Fig. 4. Analysis of the benefits of AI and BI integration

Source: Own elaboration

The analysis of barriers associated with the implementation of AI in BI systems has revealed key challenges organizations face. The study results indicate that the most frequently reported difficulties relate to technological, financial, and organizational aspects (Fig. 5). One of the most commonly identified issues is data availability and quality (20 indications). Respondents highlight challenges in integrating diverse data sources and ensuring data consistency and accuracy (18 indications). Since data forms the foundation of AI-driven BI systems, its incompleteness or misinterpretation can lead to unreliable analyses and flawed decision-making. Another critical challenge is high implementation costs (15 indications). Deploying AI-based solutions requires not only investing in appropriate software but also upgrading computational infrastructure and training employees. Many organizations struggle with budget constraints, limiting their ability to fully leverage AI's potential. A shortage of AI and BI specialists (12 indications) also emerged as a significant barrier. High competency requirements and the limited availability of experts in the job market make it difficult for organizations to recruit and retain skilled professionals capable of effectively managing and advancing AI-driven BI systems. Organizations also face integration issues between AI and existing BI

systems (11 indications). The complexity of migration processes, the need to adapt legacy technologies, and the lack of compatibility between different tools make AI implementation time-consuming and costly. Additionally, respondents frequently pointed to security and regulatory concerns (10 indications). They emphasized risks related to data protection and compliance with legal frameworks, particularly regarding GDPR and other data processing regulations.



Fig. 5. Analysis of barriers to AI and BI utilization in organizations

Source: Own elaboration

The conducted research indicates that:

1. The synergy between AI and BI in organizations leads to a significant increase in operational efficiency, improved decision-making accuracy, and better utilization of available informational resources. In the context of a rapidly changing business environment, the integration of these technologies is becoming a key determinant of strategic success for enterprises.
2. The use of artificial intelligence in BI enhances data analysis, automates processes, and improves result visualization. It enables more accurate forecasting, reduces errors, and increases operational efficiency. Its critical role in automating reporting and analytics accelerates access to relevant information. AI also supports the personalization of analyses and trend forecasting, strengthening marketing and sales strategies.
3. The main barriers to AI implementation in BI include technological limitations, high implementation costs, a shortage of highly qualified experts, integration challenges, and security and regulatory concerns. Overcoming these obstacles requires strategic investments and the development of comprehensive implementation programs that include workforce education and the standardization of data processing procedures.
4. The integration of AI and BI contributes to improved business decision quality, accelerates decision-making, and enables more effective resource management. These technologies are particularly crucial in dynamic sectors such as retail, where rapid data analysis and process optimization can provide a significant competitive advantage.

4. Conclusions and recommendations

This study explored how the integration of AI with BI enhances decision-making processes in organizations. The findings confirm that AI-enhanced BI enables a fundamental shift from retrospective reporting to proactive, data-driven decision support. Augmented Analytics- powered by machine learning and natural language processing-allows organizations to generate insights automatically, identify patterns in real time, and receive tailored decision recommendations.

In-depth interviews with managers, analysts, and technology experts revealed that AI-driven BI improves decision quality, accelerates response time, and increases forecasting accuracy. It also contributes to greater operational efficiency and supports more agile, informed strategic choices. At the same time, the implementation of AI-enhanced BI remains challenging. Key barriers identified include high implementation costs, limited access to AI specialists, and persistent issues with data quality and integration.

This research study has several limitations. The qualitative nature of the methodology, based on in-depth interviews, provides rich insights but limits its generalizability. The analysis was confined to three economic sectors: services, trade, and manufacturing - potentially reducing its relevance to other industries. Additionally, subjective interpretations from respondents may reflect personal experiences or organizational culture.

Future research should adopt mixed-methods or quantitative approaches to measure the organizational impact of AI-enhanced BI more systematically. Expanding the study to additional sectors would offer a broader perspective on adoption patterns. Further exploration of ethical and regulatory challenges - particularly related to algorithm transparency, interpretability, and data governance is also warranted. Another critical area for further investigation is the evolution of managerial and analytical competencies in organizations implementing Augmented Analytics. As AI-driven decision support systems become more widespread, the ability of managers to collaborate with intelligent systems will increasingly shape organizational success.

In conclusion, AI-enhanced Business Intelligence is no longer a technological option it is an essential driver of faster, smarter, and more strategic decision-making in the data-intensive economy.

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